

IBM SKILLSBUILD TECHNICAL PROJECT REPORT

Track: Artificial Intelligence & Agentic Orchestration Systems

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Project Title	VentureArchitect AI
Production Deployment	https://venturearchitect-ai.onrender.com
GitHub Repository	https://github.com/KARTIK-githubs/VentureArchitect-AI

1. MANDATED PROBLEM STATEMENT MAPPING

Edunet Foundation Track Requirement (Problem Statement No.20 — Startup Blueprint Generator Agent):
A Startup Blueprint Generator Agent, powered by RAG (Retrieval-Augmented Generation), transforms raw startup ideas into complete, actionable business blueprints. It retrieves relevant data on market research, funding options, competitor analysis, revenue models, government schemes, and legal requirements from startup portals, incubator databases, and policy documents. Users can describe their idea in simple terms, and the agent provides a structured plan including business model canvas, estimated budget, go-to-market strategy, and potential investor connections. Technology — Use of IBM Cloud Lite services / IBM Granite is mandatory.

2. ABSTRACT & APPLIED CORE OBJECTIVES

Translating a highly volatile early-stage startup premise into a validated enterprise asset typically requires months of fractured, manual information parsing across isolated public indexes and legal files. Traditional natural language processing models operate within flat, single-turn query constraints that lack structural continuity, introducing narrative drift and logical blind spots.

The objective of this project is to implement a production-grade Web Application that natively addresses the Edunet track requirements. By wrapping the computational capacity of enterprise AI into an asynchronous, pipeline-driven Multi-Agent blueprint engine, the application computes deterministic corporate blueprints out of unrefined technical assumptions.

3. SYSTEM ARCHITECTURE & COMPONENT WORKFLOW

The solution avoids flat chatbot behaviors by establishing a strict state orchestration pipeline. Under this model, an advanced automation manager layer receives structural variables and drives them sequentially across five independent expert microarchitectures. Each execution block preserves contextual history via schema layers (`BlueprintContext`) to achieve complete domain integrity.

- **User Request Node:** Consumes unrefined text concepts via a lightweight Flask routing layer.
- **Supervisor Orchestrator Engine:** Manages state injection, enforces sequential agent pipelines, and handles runtime validation checking.
- **AI Analytics Base Layer (IBM watsonx.ai):** Handles programmatic API text execution blocks via the enterprise-grade platform, targeting the advanced inference repository `meta-llama/llama-3-3-70b-instruct`.

4. SPECIALIZED AGENT IMPLEMENTATION BREAKDOWN

The backend engine fragments the required startup delivery criteria into five decoupled validation loops, running sequentially:

Agent Architecture Node	IBM watsonx.ai Core Responsibility	Engine Technical Outputs Generated
Idea Analysis Agent	Isolating baseline metrics and target segment configurations.	Structural problem formulation, target audience classification, value engine design.
Market Research Agent	Compiling target macro metrics and processing industry indexes.	Dynamic TAM/SAM/SOM calculation, competitive landscape profiling, threat analysis.
Business Strategy Agent	Formulating operational monetization blueprints.	Complete 9-block Business Model Canvas population, pricing tiers, operational roadmaps.
Risk Analysis Agent	Evaluating execution limits and calculating vulnerability ratings.	Risk scoring matrix, clear risk mitigations, strategic milestone scheduling.
Investor Pitch Agent	Synthesizing raw corporate telemetry data into clean narrative layouts.	60-second elevator script, 1-page corporate summary text, investor FAQ objection handling.

5. CORE NOVELTY & SOLUTION ADVANTAGES

- **State-Preserving Context Routing:** Downstream nodes receive the entire structural output trace from preceding runs, completely preventing logical conflict inside the generated strategic profile.
- **Production Server Resiliency:** Configured using robust multithreaded `Gunicorn` HTTP network wrappers on Render platforms, implementing an elongated connection window (`--timeout 300`) to handle heavy sequential inference tasks without server drops.
- **Enterprise Framework Compliance:** Natively mirrors IBM's definition of agentic AI frameworks by integrating orchestration constraints and dedicated worker topologies to execute long-lived enterprise computational workflows.

6. REPOSITORY STRUCTURE & COMPLIANCE INVENTORY

The code is structurally organized into a standard modular design package to confirm compliance requirements for public code verification:

- `app.json` : Stores environment properties and setup descriptors for automated app environments.
- `requirements.txt` : Explicitly maps package version definitions including the IBM core SDK wrapper.
- `run.py` & `app/` : The concrete implementation codebase defining the Flask services and the custom agent pipeline scripts.

7. PLANNED ENGINEERING ROADMAPS (FUTURE SCOPE)

1. **Automated Financial Engines:** Injecting targeted algorithms to handle cash flow estimations and automated break-even points.
2. **Dynamic Document Exporters:** Adding dedicated rendering engines to export generated blueprint matrices straight into professional PDF documents.
3. **Real-time Competitive Scraping:** Connecting verified live search modules to allow real-time competitor tracking and instant industry analytics updating.